

Can AI Agents Analyze Gamma-Ray Bursts?

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ABSTRACT

Yes—with a human at the gate, and the distinction matters. Analyzing a gamma-ray burst (GRB) is mostly judgement calls: which detectors to trust, where the background sits, where the burst starts and stops, how finely to slice it in time, and which of two dozen spectral models to believe. We describe a system that makes those calls, and we are precise about its architecture: it is not an autonomous agent but a *gated workflow with bounded agency*—predefined analysis paths in which an AI makes named, auditable decisions (model selection, fit validity, literature verification, mismatch attribution) while a human approves every step, and every approval is stamped. The system’s knowledge lives in a *Skill Library*: versioned plain-language procedure documents that any AI agent—or any student—can read and execute. The Library is not documentation; it is the system’s memory, and it learns. Each burst is analyzed blind, reconciled against the published literature, and every mismatch is attributed—our error, the literature’s, or a difference of frames—and distilled into a numbered lesson that ships with an executable test, so that no lesson can be silently forgotten. Applied to a campaign over 106 single-pulse *Fermi*/GBM bursts, the first ten walkthroughs^(p) produced twenty-seven such lessons, found and fixed five engine defects (one of which—a validity gate whose linear margin silently disqualified soft spectra—would have biased the census toward spurious thermal components), confirmed blind predictions drawn from the literature, and refuted one of its own. The learning curve (lessons per burst) is the paper’s central figure: the campaign trains a pipeline, not model weights, and the trained artifact is readable, testable, and transfers between AI systems. We release the Skill Library, the engine, the tests, and the per-burst records, and we report where the agent still cannot be trusted: adjudicating scientifically contested verdicts, where we show that panels of AI reviewers are strong *generators* of checkable findings but weak *adjudicators*—their unanimity can be an artifact of shared priors rather than independent measurement.

Keywords: Gamma-ray bursts (629) — Astronomy data analysis (1858) — Astronomy software (1855)
— Astrostatistics (1882)

1. INTRODUCTION

The question in our title is usually asked as if it had a yes-or-no answer. It does not, because “AI doing analysis” names at least two different architectures, and the difference between them decides what the answer can mean. In the first architecture, a *workflow*, the analysis follows predefined paths—the same stages, in the same order, for every burst—and software executes them. In the second, an *agent*, a model directs its own process: it decides what to do next, which tools to call,

and when it is finished [cite: Anthropic 2024 workflow/agent distinction; verify against the published engineering post]. Workflows are auditable and dull; agents are flexible and hard to trust. Most claims that “AI analyzed the data” blur the two.

This paper builds and tests a deliberate hybrid, and names it honestly: a *gated workflow with bounded agency*. The analysis pipeline—data acquisition, detector and background selection, adaptive binning, spectral fitting, temporal analysis, quality control—is a workflow: its stages are fixed, its statistics are chosen in advance, and its outputs are stamped with provenance. Inside that workflow, an AI exercises agency at exactly the points where judgement is required and we can af-

ford to audit it: which of twenty-four spectral models wins a gated comparison, whether a fit is valid or sits on a parameter rail, whether a published value has been verified at its source, and—when our numbers disagree with the literature—whether the fault is ours, theirs, or a difference of analysis frames. A human approves every step before the next one runs. Nothing advances on silence, and no approval is ever fabricated: every decision carries a stamp naming who, or what, made it.

Why is time-resolved GRB spectroscopy the right testbed rather than a convenient one? Because it concentrates, in one measurement problem, the three features that make scientific AI hard to evaluate. First, the decisions have no answer key. Two careful published analyses of the same burst, using overlapping data and defensible methods, can disagree about whether a thermal component exists at all; the literature is a *distribution* over defensible choices, not a ground truth [cite: Siddique et al. 2022; Dirirsa et al. 2019 – the same-burst disagreement documented in our Burst Records]. Second, the instrument matters: responses age, detectors are occulted, backgrounds are fit, and a systematic that is invisible in one regime of burst brightness or hardness appears only when the sample happens to present the other. Third, the stakes are population-level: our science application is a census of spectral shapes across 106 single-pulse bursts selected morphologically [cite: Busby & Lazzati], and a small per-burst bias becomes a population-level artifact. A system that merely *runs* on such data is easy; a system whose outputs deserve trust must show its work at every one of these pressure points.

Our design answer, developed in §3, has one unusual component that we believe is the paper’s main methodological contribution. The system’s procedural knowledge lives in a *Skill Library*: versioned, plain-language documents—one per pipeline stage—that state how the stage is run, what its known failure modes are, and what was learned from every burst that ever taught it something. The Library is executable culture: an AI agent reads the same document a new student would, and both are bound by the same rules. Crucially, the Library *learns* under a discipline we call lessons-as-tests: a lesson enters the Library only as a claim with provenance (which burst, which date, which evidence) and an executable test that fails if the lesson is ever violated again. Weights are never updated; the harness is. §4 shows what this training regime produced over the first ten bursts of the campaign, including its learning curve, and §5 develops the verification doctrine—blind-first analysis, frozen predictions, and an independence requirement we state as: agreement between two anal-

yses is evidence only in proportion to how deep their shared primitives go.

We are not the first to ask whether AI can do science end-to-end. Multi-agent systems now generate ideas, run analyses, and draft papers across disciplines [cite: Denario / Villaescusa-Navarro et al. 2025; AI-Scientist v2; verify all]. Our design differs in direction: where those systems are breadth-first—an agent writes fresh analysis code for each project—ours is depth-first: the agents operate a single, tested, domain-specific engine, and the system’s growth is the accumulation of verified domain lessons rather than of generated artifacts. The comparison is instructive in both directions, and we return to it in §8: the breadth-first systems demonstrate, at scale, exactly the evaluation gap—qualitative review standing in for ground truth—that our gated, decision-level scoring is designed to close.

The paper is organized as follows. §2 describes the sample and data. §3 presents the workflow: the Campaign Protocol, the Skill Library, the authority split between workflow, agent, and human, and the doctrine distilled from its failures. §4 presents harness learning and the campaign learning curve. §5 presents the verification doctrine and the adversarial experiments. §6 describes the human-versus-agent benchmark design. §7 reports walkthrough-era results, all provisional^(p) until a frozen production sweep. §8 discusses limits, and §9 concludes. Throughout, walkthrough-era numbers carry the flag^(p); they will be regenerated mechanically when the converged pipeline is frozen and run once over the full sample.

2. SAMPLE AND DATA

[PENDING run: port v1 §2 with the L17 inclusion-gate paragraph added]

3. THE AGENTIC WORKFLOW

3.1. The campaign loop

The unit of work is one burst, walked through eleven gated steps (Fig. ??). The *Identity & Circularity* step resolves which burst this actually is—same-day triggers exist, and we show below that this step has already intercepted a wrong redshift—and reads every real-time circular. The *Literature Harvest* step finds and fetches every refereed paper that analyzed the burst, using four query forms (name, trigger name, day-fraction, trigger number), preferring the version of record, and recording each harvested value together with its conventions: which peak energy definition, which sign convention, which energy band, which time interval on whose clock, which detectors. The *Data Inventory* step verifies that the downloaded data can support the analysis: file

versions, response-matrix time coverage, and detector geometry—including checks that no geometrically well-placed detector is physically occulted, a failure mode invisible to pointing angles. The *Selection* steps (detectors, background intervals, source interval) are made by a human in a purpose-built interface, or by a vision-capable agent, and in either case are recorded with an approval stamp naming the decision-maker and time; the pipeline treats these stamps as immutable inputs. The *Adaptive Binning* step derives Bayesian-block time bins merged to a significance floor. The *Spectral Fitting* step—where most of the agency lives—is described in §3.2. *Temporal analysis*, *spectral-energy panels*, and *quality control* complete the ledger; the QC step returns one of a set of typed terminal states, of which **non-identifiable** is a *successful* outcome: the workflow is forbidden to keep adjusting an analysis until it manufactures a preference.

Two disciplines wrap the loop. Before any comparison with the literature, the agent’s own results are frozen—written to the Burst Record in a form that cannot be revised—together with predictions derived from the harvested literature (§5). Only then are the two compared, and every mismatch is pushed to one of three attributions: *we-wrong* (an error on our side, which must be fixed and tested), *they-wrong* (a documented problem in the published analysis, stated with evidence), or *frame-difference* (both analyses defensible; the difference traced to conventions, data selections, or eras, and recorded as a normalization rule). The attribution, not the mismatch, is the product: it becomes a numbered lesson in the Skill Library, shipped with an executable test (§4).

The loop’s exit condition is convergence: when K consecutive bursts complete without producing a single new lesson, the Skill Library is declared converged, the pipeline is frozen at a recorded commit, and the entire 106-burst sample is re-analyzed once, from scratch, by the frozen system. Only that production sweep produces citable numbers; every value in this paper from the campaign era carries the provisional flag^(p). The design premise, stated plainly: *the per-burst numbers are not the product; the matured Skill Library is.*

3.2. Inside the fitting step

The fitting engine confronts each time bin with twenty-four spectral models: four one-break continua, five carrying extra low-energy structure (thermal or a second break), and fifteen high-energy composites. Three rules govern what the numbers may mean. First, *inclusion is gated on data quality, never on detection significance*: every detector band with usable data and a

valid response enters the joint fit, because conditioning inclusion on significance manufactures spurious detections in exactly the bursts where a band fluctuated upward. Second, *every model family receives unconditional multistarts*, because a chained seed from a bright epoch can strand a faint bin in a wrong local minimum—a failure we document in §4. Third, *validity gates* exclude from model selection any fit whose shape parameters sit on a bound, with the margins computed in the parameter’s natural geometry (logarithmic for energy-like parameters spanning decades)—a rule the campaign itself had to learn, at measurable cost, from its first soft burst.

Model comparison is then reported as evidence, not verdicts: the winning model’s margin is quoted both against the runner-up (*which* model) and against the best one-break continuum (*is extra structure needed at all*), each converted to an evidence ratio $e^{\Delta\text{AIC}/2}$ and graded. A margin of 6 corresponds to roughly 20:1 and is called strong; 10, roughly 150:1, decisive; anything less is reported as what it is. Where extra structure is required but thermal and non-thermal descriptions sit within a few units of each other, the bin is assigned to a degeneracy class rather than a physics claim: *adequacy and identity are different questions*, and the second frequently cannot be answered from these data alone.

3.3. Who decides what

Table ?? states the paper’s honesty contract. The *workflow* decides everything decided in advance: the step order, the energy selections, the statistic, the multistart seeds, the stamping of provenance. The *AI* decides, within audited bounds: which model wins the gated comparison, whether a fit is valid, whether a published value has been verified at source and in which frame, how a mismatch is attributed, and which anomalies deserve escalation. The *human* decides everything consequential: the scope, every step gate, whether a proposed lesson enters the Library, when convergence is declared, and every scientific claim that leaves the system. In ten bursts^(p) of operation, no approval was ever generated by the system on the human’s behalf; the reverse—the human overruling the agent’s proposal—occurred repeatedly and is part of the record (§8).

3.4. Two planes

Everything above describes the *production plane*. A second, *discovery plane* exists for ideas that do not belong inside a frozen workflow: anomalies that suggest new observables, hypotheses formed after seeing data. Work there is sandboxed and carries no evidentiary weight until it passes a promotion gate—calibration on

simulations, validation on data that did not motivate it, human approval. The governing rule is stated once and enforced always: *the agent may invent a new scorecard after seeing the game; it may not pretend the scorecard was chosen before the game.*

3.5. *The doctrine*

Seven rules, each purchased by a documented failure, govern all of the above: (1) analyze blind first—published values are harvested before fitting but the fit is never steered toward them; (2) margins are computed over valid fits only; (3) evidence ratios, never bare verdicts; (4) inclusion by data quality, never significance—and its absolute corollary: production fits never drop data that passes the quality gate, with/without refits are attribution diagnostics whose data-dropped member can never be quoted as a result, and omission of data is legal only through a documented quality exclusion; (5) a lesson is not learned until it is a claim *and* a test; (6) no silent approvals—a skipped check is a fake pass; (7) verify claims at their source, and attribute agreement or disagreement at the level of shared primitives. The failures that funded each rule appear in §4 and §8.

4. LEARNING WITHOUT GRADIENTS

[PENDING run: draft per outline §3]

5. VERIFICATION DOCTRINE

[PENDING run: draft per outline §4]

6. EXPERIMENTAL DESIGN: HUMAN VERSUS AGENT

[PENDING run: port v1 §4 + additions]

7. WALKTHROUGH-ERA RESULTS

[PENDING run: draft per outline §6]

8. DISCUSSION

[PENDING run: draft per amended outline §7 (failure modes + creativity jumps at the end)]

9. CONCLUSIONS

[PENDING run: draft last]

APPENDIX

A. REPRODUCIBILITY

[PENDING run: port v1 + fresh-clone truths + Lesson Test-Suite + release DOI]

B. ADVERSARIAL VERIFICATION PROTOCOL

[PENDING run: port v1 + feed the four-channel results]